

Communication

A Catalyst-Based Self-Sufficient System with Durable Self-Healing Functionality**

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Spontaneity and repeatability are two marked features of a biological self-healing process. Up till now, numerous synthetic self-healing materials have been designed to possess the two features. However, most of them are functional with the assistance of external energy or consumable reagents. A novel self-sufficient self-healing system is demonstrated in this work. The system is sustainable and can provide effective and durative protection for metal through autonomously healing the protective layer on metal surface while without adding any additional consumable reagents or external powers. As the system is fabricated on the basis of stable oxygen reduction reaction catalyst, the self-healing process can be repeated many times. Furthermore, the combination of catalyst-induced repeatable self-healing of metal passivation and serviceability of polymers makes this system ideal for intelligent metal protection█**Please shorten text to 700 characters maximum.█**

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Inspired by the self-healing phenomenon of biological systems in which damage triggers an autonomic healing process, materials that can heal themselves from UV radiation, mechanically, thermally, or chemically induced damages have been developed within the last decades.^[1–26] The concept of self-healing holds the expectation of prolonging lifetime and enduring performances of materials, such as mechanical strength, superhydrophobicity, and

corrosion-resistance, and so on.^[1–7] Up till now, self-healing materials for various engineering applications have been developed. Remarkably, self-healing materials for metal protection are of particular interests because developing such materials will recoup huge economic losses caused by metal corrosion, which are estimated to be 1.2–5.2% of the GDP of industrial countries.^[27]

Among various methods of metal protection, polymer coating is one of the most widely used, economical, and effective materials. Endowing polymer with self-healing functionality is a facile method to fabricate intelligent coating systems for long-term metal protection. Generally, self-healing functionality of polymer can be achieved via various approaches, including encapsulation,^[8–10] microvascular networks,^[11–15] reversible chemistry,^[16–18] polyionomers,^[19–21] nanoparticle phase separation,^[22] monomer phase separation,^[23] and so forth. And self-healing polymer can effectively maintain its performances by automatically repairing its matrix when damage occurs.^[6] However, modern-engineered coatings are highly optimized polymers in which some chemical modifications for self-healing functionality may not be acceptable in practical applications of metal protection.^[7] Besides, self-healing polymer cannot solve corrosion problems caused by inherent coating defects (such as fisheyes, craters, pinholes, blisters, and so on), which have been proved to be important causes of premature coating failure.

Over the past decade, a new self-healing system for metal protection, feedback active coating (FAC), has been proposed.^[28,29] Generally, an FAC can respond to instant or gradual impacts occurring either in the coating matrix or in the environment surrounding the coating or external signals applied to the system and release-encapsulated active materials to prevent metal from corrosion at coating defects.^[30,31] Unlike self-healing polymer, FAC prevents metal from corrosion by forming protective layers instead of repairing polymer matrix. FAC can avoid metal corrosion at inherent coating defects through directly responding to stimulations such as local pH changes, electromagnetic irradiation, electrochemical potential,

humidity, ionic strength, dielectric permeability of the solvent, and so forth.^[31] And FAC is suitable for applications in thin self-healing anticorrosive films or coatings.^[32] However, it has been demonstrated that the life-time of FAC is dominated by consumable self-healing reagents.^[33]

An ideal self-healing coating for metal protection is a sustainable self-sufficient system that can automatically restore corrosion resistance itself by utilizing available conditions in the environment surrounding coating defects (such as dissolved oxygen and production of corrosion reaction), resulting in effective and durative metal protection without adding additional powers or consumable reagents. Recent researches have revealed that it is possible to form protective layers of metal oxides at coating defects via incorporating anticorrosive pigments in the coating matrix.^[34–36] The protection phenomenon is explained by a galvanic anodic protection (GAP) mechanism, of which the kinetics of the oxygen reduction is extremely important for the durability of the protective layers.^[37–39] In this communication, we demonstrate the fabrication and characterization of a repeatable self-healing system for metal protection by using N-doped graphene/ α -MnO₂ hybrids (NGMs) as a catalyst of oxygen reduction reaction (ORR) to achieve a self-sufficient GAP for aluminum.

1. Results and Discussion

1.1. Synthesis and Characterization of NGMs

N-doped graphene (NG) was prepared by pyrolyzing well-mixed GO and urea at 800 °C in an inert environment (Figure S1 and S2). The NGMs were then synthesized by the redox reaction of NG and KMnO₄ under acidic conditions. The SEM image of the NGMs shows that uniform nanorods with a length of 217 ± 10 nm were grown on the surface of NG (**Figure 1a**). Elemental mapping results indicate that carbon, nitrogen, manganese, and oxide are

distributed evenly in NGMs (Figure 1b). As expected, a homogeneous heteroatom N-doping in the hybrids is realized. X-ray diffraction (XRD) patterns of the as-prepared NGMs show that all the marked peaks can be well indexed to the tetragonal α -MnO₂ phase (space group I4/m, PDF#44-0141) without any impurities (Figure 1c).^[40] The detailed nanostructure of NGMs is further confirmed by high-resolution transmission electron microscopy (HR-TEM), as shown in Figure 1d. The rice-like α -MnO₂ nanorods have a diameter of 28 ± 1 nm. Clear lattice fringes shown in the HR-TEM image indicate well-crystallized α -MnO₂. The chemical structure of the as-prepared NGMs is further characterized by X-ray photoelectron spectroscopy (XPS). The XPS survey spectrum also reveals the presence of C, N, Mn, and O elements in NGMs (Figure 1e). The high-resolution XPS spectrum of C 1s shows the presence of three components of carbon bonds, namely, C–C/C=C (284.9 eV), C–O (286.1 eV), and O=C–OH (288.4 eV) (Figure 1f).^[41] The O 1s spectrum of NGMs can be deconvoluted into two major bond components, corresponding to the Mn–O–C (529.9 eV) and Mn–O–Mn (531.5 eV), respectively.^[42,43] The high-resolution XPS spectrum of Mn 2p reveals that doublet peaks appear at 641.9 eV for Mn 2p_{3/2} and at 653.6 eV for Mn 2p_{1/2}, with a spin-energy separation of 11.7 eV, confirming the oxidation state of the Mn ion is +4.^[44,45] X-ray photoelectron spectroscopy (XPS) further reveals that significant contents of nitrogen ($\approx 1.1\%$ atomic percent) are incorporated into NGMs. The asymmetric N1s XPS peak of NGMs can be fitted to three peaks of pyridinic N (398.3 eV), pyrrolic N (400.1 eV), and graphitic N (401.3 eV).^[46]

1.2. Electrocatalytic Performance

To automatically offer GAP for aluminum, NGMs should not only offer immediately a large initial current to induce metal passivation once the coating is damaged, but also provide a steady-state current that is enough to maintain metal passivation.^[47,48] For a sustainable self-

sufficient system of GAP, the GAP performance of NGMs-incorporated coating depends on the ORR catalytic activity of NGMs because GAP-required current could be generated by the ORR catalyzed by NGMs (or galvanic current of NGMs–metal micro-couple). The cyclic voltammogram (CV) for oxygen reduction on the NGMs electrode in an O₂-saturated 3.5 wt% NaCl aqueous solution is shown in **Figure 2a**. It can be seen that the onset potential of ORR for the NGMs electrode is at −0.28 V versus SCE with the weak cathodic reduction peak around −0.45 V versus SCE, indicating the catalytic activity of the NGMs. To further investigate the ORR performance, the linear sweep voltammetric (LSV) measurements on a rotating disk electrode (RDE) with NGMs were carried out in an O₂-saturated 3.5 wt% NaCl solution, as evidenced in Figure 2b. The diffusion current density enhances with the increase of rotation rate from 400 to 2 025 rpm due to the enhanced mass transfer in the electrolyte. Figure 2b also displays the Koutecky–Levich plots of NGMs electrode at −0.7, −0.75, and −0.8 V versus SCE, respectively. And these Koutecky–Levich plots exhibit a good linearity and parallelism, which is an indication of first-order reaction kinetics with respect to the concentration of dissolved O₂.^[49,50] The LSV results of NGMs and α -MnO₂ show that the electrocatalytic current varied strongly with the measured potential ranging from −1.2 to −0.6 V versus SCE, while that of NG changes slowly (Figure S3). The ORR catalytic activity of NGMs needs to be sensitive to the change in potential in an autonomous GAP system because the current required at each stage of anodic protection varies widely. Therefore, NGMs and α -MnO₂ are better catalysts than NG for a sustainable self-sufficient GAP system. Furthermore, four-electron ORR is more conducive to GAP than two-electron ORR because the hydroxide ions produced by four-electron ORR can not only expel aggressive chloride ions from the scratch via electrostatic repulsive force, but also create a weak alkaline environment (pH < 10) for maintaining the passivation film;^[51,52] while H₂O₂ generated by two-electron ORR tends to destroy the passivation film.^[53,54] In this work, a rotating ring–disk electrode (RRDE) technique was employed to accurately determine the amount of H₂O₂ generated at the disk

electrode and the electron transfer pathway (n) of ORR. The inset in Figure 2c shows the disk and ring currents for NGMs, α -MnO₂, and NG, respectively. All three electrodes generated ring currents at the onset potential for the ORR. Remarkably, both α -MnO₂ and NG exhibited much higher ring currents than NGMs. Figure 2c reveals that the electron transfer number varied a little with the measured potential. On the basis of the ring and disk currents, the electron transfer numbers were 2.99–3.24 for NG and 3.67–3.76 for α -MnO₂ over the potential range from –0.40 to –1.2 V versus SCE. In sharp contrast, the $n = 3.88$ –3.93 for NGMs electrode over the potential ranging from –0.40 to –1.2 V versus SCE. Furthermore, the measured H₂O₂ yield for NGMs was 3.0–5.6%, while those for α -MnO₂ and NG were 12–16 and 38–51%, respectively (Figure 2d). RDE and RRDE study emphasizes that the NGMs ORR proceeds mainly via a desired four-electron mechanism.^[55] Hence, it is deduced that non-consumable NGMs are more suitable catalysts for repeatable GAP, and this deduce was further verified by electrochemical noise analysis (ENA) and electrochemical impedance spectroscopy (EIS).

1.3. Self-Healing Performance of NGMs-Embedded Coating

The synthesized NGMs were incorporated into polyvinyl butyral (PVB) coating, and then the composite coating was coated onto an aluminum substrate (the obtained specimen was denoted as NGMs/PVBc). The NGMs/PVBc was artificially scratched and immersed in 3.5 wt% NaCl solution. In order to study the self-healing performance of NGMs/PVBc, ENA was used to monitor continuously the corrosion behavior of aluminum and the formation process of the protective layer. Potential noise (shown in Figure S4) was analyzed by wavelet technique based on orthogonal sym4 wavelet (a typical wavelet analysis of the potential noise is shown in Figure S5). The characteristics of energy distribution plot (EDP) obtained from wavelet analysis technique are “fingerprints” of the noise signal and can be used to

differentiate the reaction type at the metal/electrolyte interface.^[56,57] Generally, it is accepted that the initiation, growth, and death of metastable process always occur prior to corrosion, whereas corrosion is faster than other processes such as diffusion of aggressive ions or oxygen molecules.^[58] If the time-scale of different crystal D was ranked in an increasing order: $D_1 < D_2 < \dots < D_7 < D_8$, the relative energy distribution in crystal D_1 – D_2 , D_3 – D_6 , and

D_7 – D_8 would mainly reflect the information about metastable processes, corrosion, and diffusion, respectively.^[59,60] The time-dependent evolution of EDP corresponding to potential noise of scratched PVBC and NGMs/PVBC during a short immersion is shown in **Figure 3**. The EDP of PVBC shows a random energy distribution between D_3 and D_8 , indicating that there are aluminum corrosion and oxygen diffusion process near the aluminum/electrolyte interface of damaged PVBC (Figure 3a).^[61] The corrosion of PVBC is further confirmed by an SEM image of the scratch, in which obvious fluffy corrosion products are observed (Figure 3b). However, the EDP of scratched NGMs/PVBC reveals that the energy distributes at slow process region D_7 – D_8 during 0–10 h immersion and fast process region D_1 – D_2 during 10–45 h immersion (Figure 3c). The slow process is attributed to the oxygen diffusion caused by serious corrosion of the aluminum substrate, and the fast process corresponds to a metastable process raised by the dissolution/formation of protective layer (aluminum passivation).^[62] No obvious localized or general corrosion information is detected by ENA technology, indicating that the NGMs/PVBC is being protected well. This conclusion is further confirmed by an SEM image of the scratch which shows little corrosion product (Figure 3d). The potential noise was also analyzed by fast Fourier transform and dimensionless analysis. The analysis results reveal that the exposed aluminum surface firstly underwent a slow process during initial 15 h

immersion and subsequently a relatively fast process during 15–45 h immersion (Figure S6). The fast Fourier transform and dimensionless analysis agree well with that obtained from the wavelet analysis, confirming the self-healing functionality of NGMs/PVBc. The self-healing functionality is believed to be related to the ORR occurred on NGMs because, on the one hand, only Al and O element can be detected in the damaged region on aluminum (Figure S7) and, on the other hand, the self-healing phenomenon cannot be observed in deoxygenized NaCl solution (Figure S8). For comparison, the corrosion behavior of α -MnO₂/PVBc and NG/PVBc was also investigated by ENA. The evolution of the potential noise of α -MnO₂/PVBc is similar to that of NGMs/PVBc. The corresponding wavelet analysis result shows that the energy distribution range changes from D_4 – D_8 to D_1 – D_4 . The potential noise of NG shows a violent fluctuation throughout the whole experiments, and the wavelet analysis result of the potential noise exhibits a strong random energy distribution between D_1 and D_8 (Figure S9). The ENA results also reveal a weak self-healing performance of α -MnO₂/PVBc and a corrosion-promotion activity of NG/PVBc.

EIS was measured at different immersion times to further evaluate the self-healing performance of NGMs/PVBc. The evolution of the Bode-phase plots of NGMs/PVBc reveals a process of peak splitting. Initially, only one obvious peak locating at frequencies ranging from 10^4 to 10^0 Hz can be observed (**Figure 4a**). Subsequently, a peak gradually appears at low frequencies ranging from 10^1 to 10^{-2} Hz, and another peak gradually appears at high frequencies ranging from 10^5 to 10^1 Hz. After being immersed for 24 h, the Bode-phase plot remains essentially unchanged, indicating that the system reached a stable state. Generally, for Bode-phase plot with a plurality of peaks, the peak in high-frequency region is attributed to the responses of protective film and that appearing at medium/low frequency corresponds to the responses of metal corrosion.^[24–26] Figure 4b shows that the modulus of NGMs/PVBc gradually increases during the early 24 h immersion. Especially, the modulus increases rapidly at low frequencies. Therefore, it is concluded that a protective film is formed on the

aluminum surface at the defects. However, the Bode-phase plot of aluminum coated by blank PVB coating (the obtained specimen was denoted as PVBC) shows an obvious peak located at frequencies ranging from 10^2 to 10^{-2} Hz which is attributed to the corrosion of aluminum at the defects and a weak peak at frequencies ranging from 10^4 to 10^2 Hz (Figure 4c), ascribing to the formation of porous corrosion products on the surface of aluminum at the defects (Figure 3b). During the immersion process, the modulus of PVBC changes little (Figure 4d). The Zsimpwin software was used to quantitatively analyze the EIS results using the equivalent circuit model shown in Figure S10. The R_{total} profile of PVBC exhibits a continuous fluctuation in the range of 10^5 – 10^6 Ω throughout the immersion process (Figure 4e). While, the total resistance (R_{total}) of the NGMs/PVBC increases from 1.53×10^6 to 1.80×10^7 Ω during the first day immersion, and subsequently maintains above 10^7 Ω . The R_{total} of NGMs/PVBC is about 1–2 orders of magnitude larger than that of PVBC, indicating NGMs/PVBC possesses a good self-healing functionality that can restore the corrosion-resistance of aluminum. Besides, NGMs/PVBC is able to sustainably and stably generate protective current to maintain the protective layer for a long-term period. The self-healing performances of α -MnO₂/PVBC and NG/PVBC were also investigated in this work. Similarly with NGMs/PVBC, α -MnO₂/PVBC exhibits a weak passivity peak at high frequencies in the Bode-phase plots (10^5 – 10^3 Hz), and its R_{total} is about an order of magnitude larger than that of PVBC (Figure S11a–c), indicating that the α -MnO₂/PVBC possesses a weak self-healing performance. While for NG/PVBC, EIS results reveal that incorporated NG accelerates rather than inhibits aluminum corrosion (Figure S11d–f).

To verify the repeatable self-healing functionality of the sustainable self-sufficient GAP system, NGMs/PVBC was carefully scratched every 24 h at the same site to repeatedly destroy the protective layer. And the EIS was used to further monitor the variation tendency of the impedance spectra of the damaged NGMs/PVBC

in 3.5 wt% NaCl solution. The R_{total} of NGMs/PVBc has an initial value of $\approx 2.51 \times 10^6 \Omega$ after the first damage. As the immersion time increases, the R_{total} of NGMs/PVBc gradually increases. After 24 h immersion, the R_{total} reached $1.58 \times 10^7 \Omega$, which results in a suppression of the corrosion at the site of the scratch (Figure 4f). When damaging the protective layer, the R_{total} promptly decreased to $1.01 \times 10^6 \Omega$, which indicates the occurrence of corrosion process. Subsequently, the corrosion activity disappears after another 24 h immersion, since the R_{total} gradually increased to $1.40 \times 10^7 \Omega$. When repeating the same operation, the self-healing phenomenon of the R_{total} reappeared. Therefore, such a self-healing phenomenon is self-sufficient and repeatable without additional powers (e.g., electric power, UV irradiation) or reagents (e.g., corrosion inhibitor).

2. Conclusions

We demonstrated a strategy for the fabrication of a sustainable self-healing system based on non-consumable ORR catalysts for metal protection. The developed system has an excellent self-healing capability which can repeatedly and effectively restore the protection barrier of metals via an autonomous mechanism without any additional powers, consumable reagents, or external stimulus. This intrinsic self-healing functionality is sustainably self-sufficient. And the self-healing is explicated by a GAP mechanism, which states that the initiating passivation current and maintaining passivity current are provided by the ORR occurred on the catalysts. Furthermore, ORR which proceeds via four-electron pathway is conducive to create a suitable environment for the maintenance of protective layers. The concept of sustainable self-sufficient self-healing systems is able to provide long-term protection for metals even when the cured defect is damaged again. Such an ORR catalyst-based self-healing system offers a simple and environment-friendly strategy for the intelligent protection of metals.

3. Experimental Section

3.1. Synthesis of Graphene Oxide

Graphene oxide (GO) was synthesized from natural graphite flakes by Hummers' method. Typically, H_2SO_4 (46 mL, 98 wt%) was added into a flask, and then cooled in an ice bath. Subsequently, graphite powder (2.0 g) and NaNO_3 (1.0 g) were successively added under stirring. Under vigorous stirring, KMnO_4 (6.0 g) was added slowly over 5 min to avoid overheating. The mixture was maintained at 0 °C for 30 min. Successively, the reaction system was transferred to a 35 °C water bath and stirred for another 30 min, forming a thick paste. Then, deionized water (100 mL) was added slowing (controlling the mixture did not boil), and then the solution was transferred to a 98 °C water bath for 30 min. Deionized water (100 mL) was added to dilute the solution. Finally, H_2O_2 (30 wt%) was added into the solution, and the slurry immediately turned into a bright yellow solution with bubbling, followed by centrifugation and careful washing to clean out remnant salts.

3.2. Synthesis of N-Doped GO

The obtained GO paste was dispersed in deionized water (200 mL) by stirring for 30 min. Then, urea was (40 g) added into the GO suspension, and the mixture was continuously stirred for 30 min to completely dissolve the urea. The solution was dried in a rotary evaporator at 60 °C for 2 days. The obtained GO/urea mixture was pyrolyzed at 800 °C for 30 min to produce NG.

3.3. Synthesis of NGMs

NG (0.2 g) was firstly dispersed in sulfuric acid solution (400 mL 2 M) at 90 °C for 2 h with stirring. Then, KMnO_4 (2 g) was added into the solution and the mixture was maintained at 90 °C with stirring for 30 min and was cooled to room temperature naturally after the reaction was finished. The precipitation was filtered and washed with de-ionized water and ethanol repeatedly and then dried at 85 °C overnight. Finally, the obtained powder was calcined at 500 °C for 1 h.

3.4. Preparation of NGMs/PVBc

NGMs (0.1 g) were completely dispersed in methanol (20 mL) in an ultrasonic bath. Then, polyvinylbutyral (2 g, PVB) powders were slowly added to the dispersion. A viscous NGMs/PVBc solution was obtained by continuously stirring the mixture for 1 day. Before coating, aluminum substrate ($2 \times 4 \text{ cm}^2$) was ground with 1 000 SiC grit papers, and ultrasonically cleaned in acetone for 5 min. The coating solution was dip-coated on the pretreated Al substrate, and then dried for 1 day at 20 °C to obtain the NGMs/PVB-coated Al substrate. Similarly, $\alpha\text{-MnO}_2$ (0.1 g) and NG (0.1 g) were used to prepare the contrast specimens $\alpha\text{-MnO}_2/\text{PVB}$ and NG/PVB, respectively. For each specimen, the coating thickness was controlled to be $82.4 \pm 3.5 \text{ }\mu\text{m}$ (TR110, Beijing Time High Technology Ltd., China).

3.5. Characterizations

The morphology and crystal structure of the NGMs were characterized by X-ray scanning electron microscopy (SEM, Zeiss EVO LS 15, Germany), transmission electron microscopy (TEM, FEI Tecnai F30, Holland), and diffraction (XRD, Shimadzu XRD-6000, Japan), respectively. XPS analysis was performed by using an ESCALAB 250 instrument with Al KR radiation (15 kV, 150 W) under a pressure of $4 \times 10^{-8} \text{ Pa}$.

3.6. Electrochemical Measurements

The electrochemical properties of the NG were tested using a conventional three-electrode system by an electrochemical workstation (CHI 760D, Shanghai, China). A Pt fossil and an Hg/Hg₂Cl₂ electrode filled with saturated KCl aqueous solution were used as the counter electrode and reference electrode, respectively. To prepare the NGMs-loaded working electrode, the NG was dispersed in a mixed solution consisting of water, ethanol, and Nafion (5 wt%, Dupont) ($V_{\text{water}}:V_{\text{ethanol}}:V_{\text{Nafion}} = 1:9:0.1$) by sonication. The NGMs dispersion (2 mg mL⁻¹) was transferred onto a glassy-carbon (GC) electrode (0.19625 cm², PINE Instruments) and dried at room temperature. The NGMs loading was calculated to be 0.2 mg cm⁻². The control-sample NG and α -MnO₂ on GC were prepared in the same way.

The cyclic voltammetry (CV) and linear-scan-voltammetry (LSV) measurements were also performed using the workstation. The CV characterization of catalysts was carried out in N₂-purged 3.5 wt% NaCl aqueous solution with a potential range of -1.2 to 0.2 V versus SCE at a positive scanning rate of 50 mV s⁻¹. The working electrode was cycled at least five times before data were recorded. The LVS curves were obtained at the potential ranging from -1.2 to 0.2 V versus SCE at a positive scanning rate of 10 mV s⁻¹.

The oxygen reduction reaction (ORR) polarization curves of RDE and RRDE (disk area: 0.2475 cm², ring area: 0.1886 cm², PINE Instrument) were obtained in O₂-saturated 3.5 wt% solution with a scanning range of -1.2 to 0.2 V versus SCE and a positive scanning rate of 10 mV s⁻¹ at 1 600 rpm. The loading amount of ORR catalyst was controlled to be 0.2 mg cm⁻². The ring potential of RRDE was kept constant at 0.2 V versus SCE. The peroxide percentage (%HO₂⁻) and electron transfer number (n) were evaluated based on the following equations:

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(2)

(3)

In Equation 1, J_{lim} is the limiting current density; n is the number of electrons transferred per oxygen molecule; F ($96\,485\text{ C mol}^{-1}$) is the Faraday constant; D ($1.9 \times 10^{-5}\text{ cm}^2\text{ s}^{-1}$) is the diffusion coefficient of O_2 in 3.5 wt% NaCl solution, and C_0 ($1.2 \times 10^{-6}\text{ mol cm}^{-3}$) is the concentration of O_2 in the electrolyte; ν is the kinetic viscosity of the solution ($0.01\text{ cm}^2\text{ s}^{-1}$); and ω is the electrode rotation rate (rad s^{-1}). In Equation 2 and 3, I_r and I_d are the ring and disk current, respectively. The manufacturer's value of N (ring collection efficiency) is 37%.

The Koutechy–Levich equation can also be applied to determine the electron transfer number of per oxygen molecule involved in the ORR process occurring on the MNGs electrode.

(4)

(5)

where J is the measured current density, B is the Levich constant, ω is the angular velocity, J_k is the kinetic current density.

Electrochemical impedance spectroscopy (EIS) measurements were performed on a CHI660D electrochemical workstation (Chenhua, China) with a three-electrode system. A platinum wire was used as auxiliary electrode, a saturated calomel electrode (SCE) as reference electrode, and the aluminum as working electrode. The experimental solution was 3.5 wt% NaCl. A sinusoidal AC perturbation of 5 mV (rms) amplitude coupled with the corrosion potential of work electrodes is then applied to the electrode over the frequency range 0.01–100 kHz. For EIS, all data analysis was performed using ZsimpWin software.

Electrochemical noise (EN) was measured in a freely corroding system using two nominally identical working electrodes of the same area and a saturated $\text{Hg}/\text{Hg}_2\text{Cl}_2$ reference

electrode which was placed between the two working electrodes. During the electrochemical measurements, the cell was placed in a Faradic cage to minimize possible external electromagnetic interference. The electrochemical noise data were gathered within a period of 45 h. The sampling frequency was 2 Hz. EN measurements were carried out with a CS300 workstation containing a noise module (CorrTest, Wuhan, China). For EN, all data analysis was performed using Matlab R2012b software.

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Figure caption

Fig. 1. Morphology and structure of the NGMs. (a) Typical SEM images of NGMs revealing the 3D structure and uniform distribution of α -MnO₂ nanorods on the NG. (b) Elemental mapping, (c) XRD pattern, and (d) HR-TEM of the NGMs. (e) Survey XPS spectrum of the NGMs. (f) High-resolution XPS spectra of C 1s, N 1s, Mn 2p, and O 1s.

Fig. 2. (a) CV curves (scanning rate of 50 mV s⁻¹) of the NGMs on a glassy-carbon RDE in a N₂- and an O₂-saturated 3.5 wt% NaCl solution. (b) LSVs (10 mV s⁻¹) for the NGMs on an RDE with various rotating speeds. Inset in (b) is corresponding Koutecky–Levich plots at different potentials. (c) Electron transfer number and (d) peroxide percentage of NGMs as functions of the electrode potential. Inset (c) is RRDE test of the ORR on NGMs, α -MnO₂, and NG in an O₂-saturated 3.5 wt% NaCl solution at a rotation rate of 1 600 rpm.

Fig. 3. Contour plot of the energy distribution of scratched (a) pristine PVBc and (c) NGM/PVBc. SEM image of the scratch of (b) PVBc and (d) NGM/PVBc-coated aluminum.

Fig. 4. (a) The modulus and (b) phase Bode plot of NGMs/PVBc. (c) The modulus and (d) phase Bode plot of pure PVBc. (e) The total resistance evolution of scratched NGMs/PVBc and PVBc. The average total resistance of at least three different samples is presented, and the error bars represent the standard deviations. (f) The total resistance evolution of a repeatedly scratched specimen coated with NGMs/PVBc.

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